Spring Research Project: Mentor Rubric

Mentor Name(s): [redacted]

Mentee Name(s): [redacted]

Project Title: Exoplanet detection using the RV method

In a short paragraph, explain the project and its learning objectives:

In this project, the mentee will learn about different methods of exoplanet detection. For this project, the focus is on the radial velocity (RV) method. The mentee will learn python skills on data analysis (working with data files, plotting). The mentee will also learn how to use the RV method to determine properties of an exoplanet orbiting its host star.

What are the deliverables? (i.e. plot/figure/image/equation mentee will reach)

The mentee will create a radial velocity plot, along with values for the planet's period, semi-major axis, and mass. The final presentation will include this, as well as information about the radial velocity method, the equations used to determine the period, semi-major axis, and mass of the planet. The project can be expanded to determine these quantities for multiple systems.

POLARIS

Project Calendar: Indicate what the mentee will be expected to do each week. Work days are indicated in the left column. Include estimated time, resources, in/out of class work, and whether it is completed on their own or with you. The project should take 8-10 hours in total.

Dates:	Tasks:
Week 1 (2/10-2/16) Intro to Project	Outside of class: background reading, radial velocity simulator + worksheet (questions 1-10) (estimated time: 1hr)
Week 2 (2/17-2/23) In Class Work Day	In class: discuss background readings + simulator + worksheet Outside of class: python introduction (mentor will give tutorial notebooks) (estimated time: 1.5hr)
Week 3 (2/24-3/2) In Class Work Day	In class: discuss python, provide skeleton Jupyter Notebook for the project and discuss how to work with it Outside of class: work with RV data in python (plot data + extract properties of the planet) (estimated time: 1hr)
Week 4 (3/3-3/8) Class Lecture	Outside of class: working with RV data, mentor will give more data files if mentee is interested (estimated time: 1hr)
Spring Break (3/10-3/14)	None
Week 5 (3/17-3/23) In Class Work Day	In class: discuss results/issues Outside of class: continue working with RV data, mentor will give more data files if mentee is interested (estimated time: 1hr)
Week 6 (3/24-3/30) Class Lecture	Outside of class: finalize plot(s), summary on what was learned (estimated time: 1hr)
Week 7 (3/31-4/6) Class Lecture	Outside of class: rough draft of poster (estimated time: 1hr)
Week 8 (4/7-4/13) In Class Work Day	In class: go over summary and poster Outside of class: finalize poster (estimated time: 1hr)
Week 9 (4/14-4/20) Poster Work Day	In class: go over poster, practice presentation Outside of class: practice presentation (estimated time: 30min)

POLARIS

Week 10 (4/21) Present!	

Reading List: (Make sure material is level appropriate.)

- NAAP Lab: https://astro.unl.edu/naap/esp/esp.html
  - Readings listed under “Main Content”
  - Questions 1-10 of “Student Guide” + “Exoplanet Radial Velocity Simulator”
- Mentor will provide python notebook tutorials
  - Python intro
  - Matplotlib intro
  - Curve fit intro (optional/reference)
- Other code resources: https://prappleizer.github.io/index.html (optional/reference)

Resource List: (This should include lab space, instruments, and software. Note that mentees are not allowed to complete research projects that require safety training operation of lab equipment. If you feel this is detrimental to your project, please petition Polaris leadership.)

- Laptop with internet access
- SciServer account (free; https://www.sciserver.org/)

Other Notes:

This project can be expanded to analyze radial velocity curves for multiple systems. The mentor will provide data files.